

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Hackathon	Assessment:	Assessment Tool 3– Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and interpret datasets using data preprocessing, statistical analysis, visualization, and machine learning techniques for effective data-driven insights and decision-making.		
Student Name:	V.Sai Charan	Reg. No.:	192472159
Section / Batch:	CSE – AI	Date:	29/08/26

Code of Conduct

I V.Sai Charan– 192472159 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Sai Charan

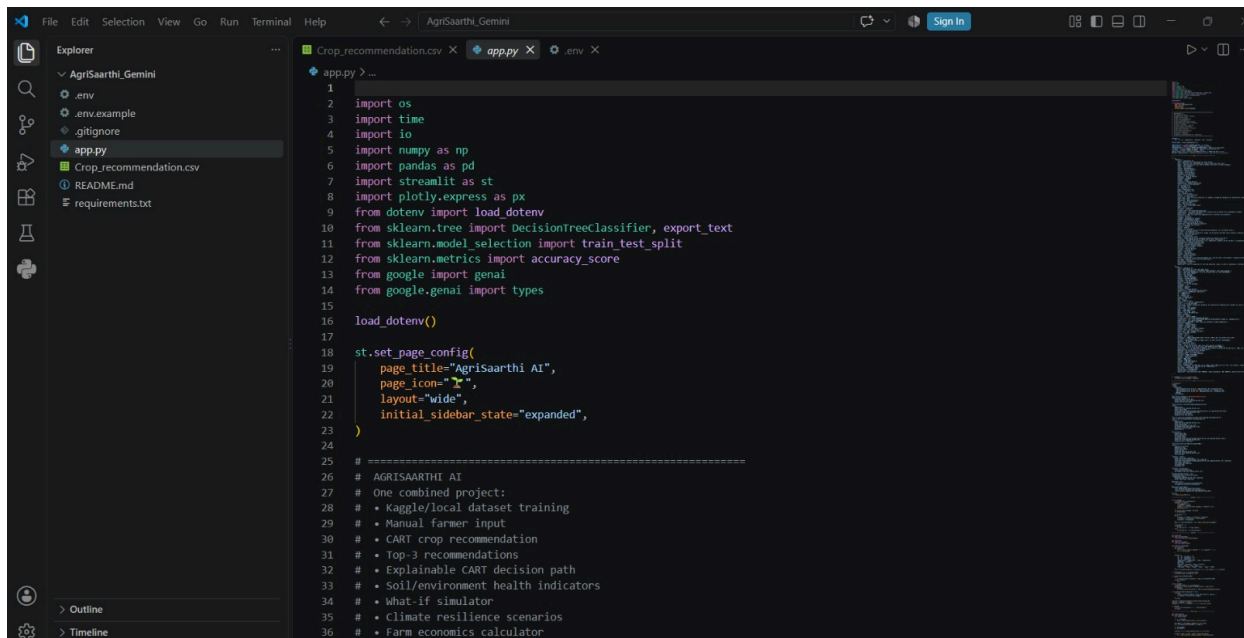
Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	I n - d e p t h analysis with s t r o n g reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandab le	Limited clarity and structure	Poor clarity; difficult to understand	

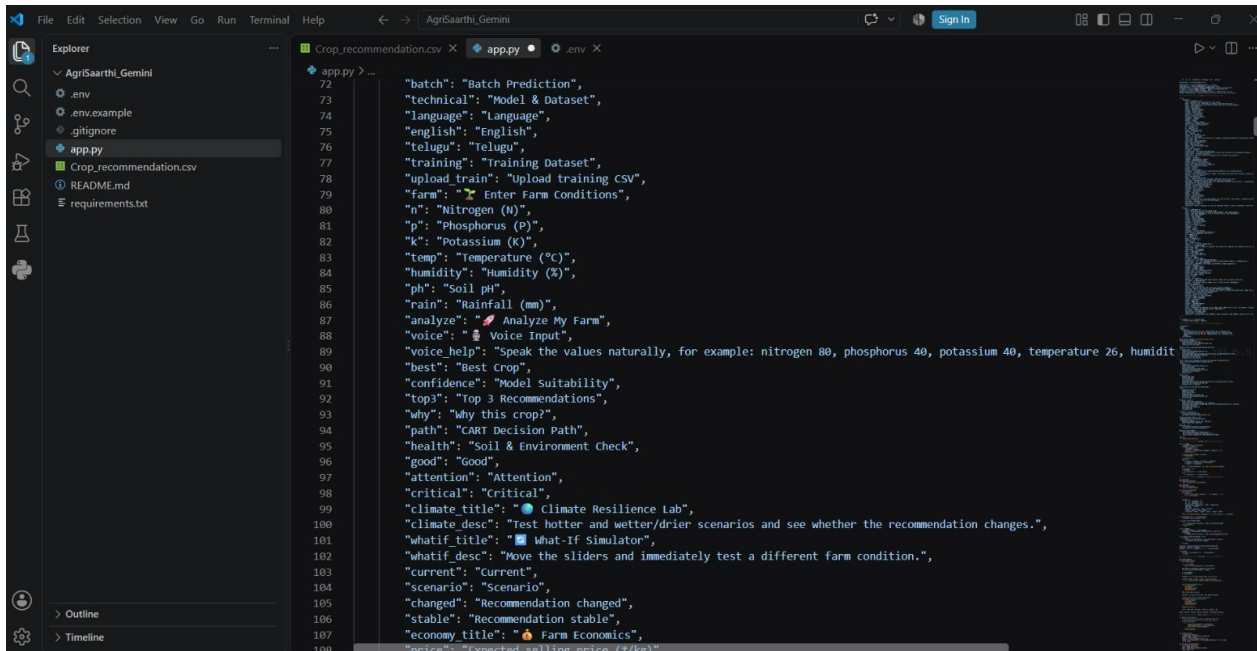
DSA0405 Fundamentals of Data Science

CO5-AT3 – Hackathon

Code:



```
1
2 import os
3 import time
4 import io
5 import numpy as np
6 import pandas as pd
7 import streamlit as st
8 import plotly.express as px
9 from dotenv import load_dotenv
10 from sklearn.tree import DecisionTreeClassifier, export_text
11 from sklearn.model_selection import train_test_split
12 from sklearn.metrics import accuracy_score
13 from google import genai
14 from google.genai import types
15
16 load_dotenv()
17
18 st.set_page_config(
19     page_title="AgrisSaarthi AI",
20     page_icon="🌾",
21     layout="wide",
22     initial_sidebar_state="expanded",
23 )
24
25 # =====
26 # AGRISAAATHI AI
27 # One combined project:
28 # * Kaggle/local dataset training
29 # * Manual farmer input
30 # * CART crop recommendation
31 # * Top-3 recommendations
32 # * Explainable CART decision path
33 # * Soil/environment health indicators
34 # * What-if simulator
35 # * Climate resilience scenarios
36 # * Farm economics calculator
```



```
72 "batch": "Batch Prediction",
73 "technical": "Model & Dataset",
74 "language": "Language",
75 "english": "English",
76 "telugu": "Telugu",
77 "training": "Training Dataset",
78 "upload_train": "Upload training CSV",
79 "farm": "🌾 Enter Farm Conditions",
80 "n": "Nitrogen (N)",
81 "p": "Phosphorus (P)",
82 "k": "Potassium (K)",
83 "temp": "Temperature (°C)",
84 "humidity": "Humidity (%)",
85 "ph": "Soil pH",
86 "rain": "Rainfall (mm)",
87 "analyze": "🔍 Analyze My Farm",
88 "voice": "🗣️ Voice Input",
89 "voice_help": "Speak the values naturally, for example: nitrogen 80, phosphorus 40, potassium 40, temperature 26, humidity 60",
90 "best": "Best Crop",
91 "confidence": "Model Suitability",
92 "top3": "Top 3 Recommendations",
93 "why": "Why this crop?",
94 "path": "CART Decision Path",
95 "health": "Soil & Environment Check",
96 "good": "Good",
97 "attention": "Attention",
98 "critical": "Critical",
99 "climate_title": "🌍 Climate Resilience Lab",
100 "climate_desc": "Test hotter and wetter/drier scenarios and see whether the recommendation changes.",
101 "whatif_title": "📊 What-if Simulator",
102 "whatif_desc": "Move the sliders and immediately test a different farm condition.",
103 "current": "Current",
104 "scenario": "Scenario",
105 "changed": "Recommendation changed",
106 "stable": "Recommendation stable",
107 "economy_title": "💰 Farm Economics",
108 "noise": "Expected selling price (₹/kg)"
```

Implementation:

AgriSaarthi

Language / భాష

English Telugu

హిలం డ్యాష్‌బోర్డ్

హిలం వివరాలు

పంట సిఫార్సు

ఏమైతే సిమ్ములేటర్

వాతావరణ ప్రయోగశాల

హిలం ఆర్థిక లెక్కలు

AI సహాయకుడు

బ్యాచ్ అంచనా

మోడల్ & డేటాసెట్

API సిద్ధంగా ఉంది

Training Dataset

Training CSV అప్‌లోడ్ చేయండి

ప్రతి రైతుకు AI ఆధారిత వంట నిర్ణయ సహాయం

అగ్రిసార్థి AI

మీ నేలను తెలుసుకోండి. వాతావరణాన్ని అర్థం చేసుకోండి. సరైన పంటను ఎంచుకోండి.
CART పంటను సూచిస్తుంది. నిజమైన AI ఫలితాన్ని సులభమైన భాషలో వివరిస్తుంది.

CARTKaggle DatasetGeminiClimate LabEconomicsEnglish + తెలుగు

హిలం డ్యాష్‌బోర్డ్

73.9%వరీక్ష ఖచ్చితత్వం

2200శిక్షణ రికార్డులు

22పంట రకాలు

7మోడల్ & డేటాసెట్

Complete farmer workflow

AgriSaarthi

Language / భాష

English Telugu

Farm Dashboard

Farm Inputs

Recommendation

What-If Simulator

Climate Lab

Farm Economics

AI Assistant

Batch Prediction

Model & Dataset

API Ready

Training Dataset

Upload training CSV

AI-powered crop intelligence for every farmer

AgriSaarthi AI

Know your soil. Understand your climate. Choose the right crop.
CART predicts the crop. Real AI explains the result in simple language.

CARTKaggle DatasetGeminiClimate LabEconomicsEnglish + తెలుగు

Enter Farm Conditions

Nitrogen (N)

30.00

Temperature (°C)

26.00

Rainfall (mm)

180.00

Phosphorus (P)

37.00

Humidity (%)

74.00

Voice Input

Record

00:00

Potassium (K)

36.00

Soil pH

6.50

Speak the values naturally, for example: nitrogen 80, phosphorus 40, potassium 40, temperature 26, humidity 70, pH 6.5, rainfall 180.

AgriSaarthi

Language / భాష

English Telugu

Farm Dashboard

Farm Inputs

Recommendation

What-If Simulator

Climate Lab

Farm Economics

AI Assistant

Batch Prediction

Model & Dataset

API Ready

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Upload training CSV

Deploy

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CARTKaggle DatasetGeminiClimate LabEconomicsEnglish + తెలుగు

Mungbean
Best Crop

100.0%
Model Suitability

Good
Soil & Environment Check

Top 3 Recommendations

AgriSaarthi

Language / భాష

English Telugu

Farm Dashboard

Farm Inputs

Recommendation

What-If Simulator

Climate Lab

Farm Economics

AI Assistant

Batch Prediction

Model & Dataset

API Ready

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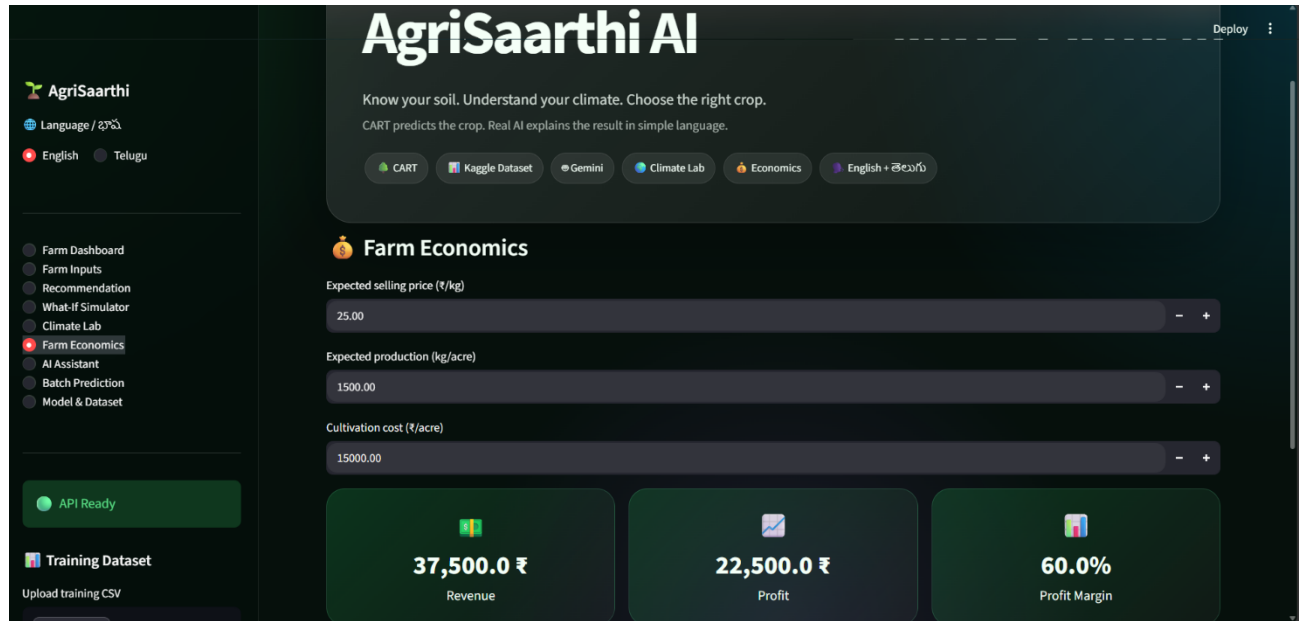
AgriSaarthi AI

Ask a question in English or Telugu. The AI explains the the CART result instead of replacing it.

Ask your question

Why did you recommend this crop?
ఈ ఫంటు ఎందుకు సిఫార్సు చేశారు?

Ask AgriSaarthi



Geotag:

